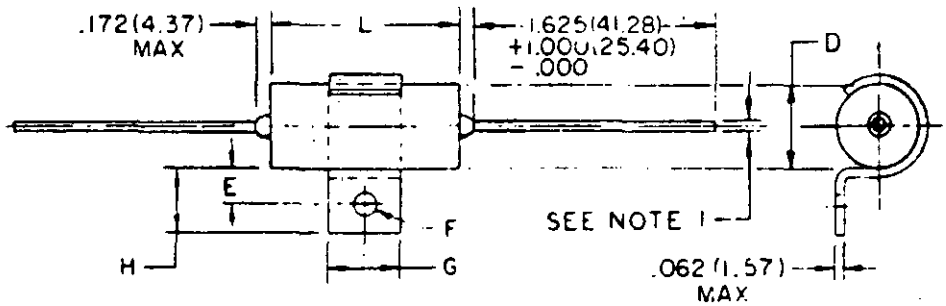
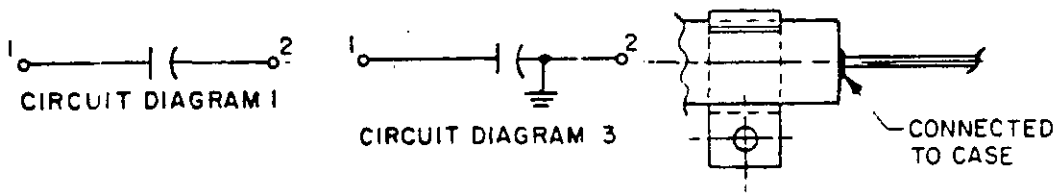
MILITARY SPECIFICATION SHEET

CAPACITORS, FIXED, METALLIZED, PAPER-PLASTIC FILM DIELECTRIC, DIRECT CURRENT, (HERMETICALLY SEALED IN METAL CASES), ESTABLISHED RELIABILITY, STYLE CHR12 (UNINSULATED)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the capacitors described herein shall consist of this document and the latest issue of Specification MIL-C-39022.





NOTES:
1. Number 22 AWG wire for case diameters of .312 (7.92 mm).
   Number 20 AWG wire for case diameters over .312 (7.92 mm).
2. See table I for additional dimensions.
3. Dimensions are in inches.
4. Metric equivalents (to the nearest .01 mm) are given for general information only, and are based upon 1 inch = 25.4 mm.
5. Metric equivalents are in parentheses.

Ⓑ denotes changes.

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REQUIREMENTS:

Design and construction:

Dimensions and configuration: See figure and table I.

Case material: Nonmagnetic.

Capacitance value: See table I.

(B) Capacitance tolerance: See table I.

Rated voltage: See table I.

Dielectric material:
Normally paper-polyethylene-terephthalate (on 200-, 400-, and 600-volt units).
(B) Normally polyethylene-terephthalate (on 50-volt units).

Operating temperature range:
-55° to +125°C (on 200-, 400-, and 600-volt units).
(B) -55° to + 85°C (on 50-volt units).

Failure rate level: L(5.0%), M(1.0%), P(0.1%), R(0.01%), and S(0.001%).

Radiographic inspection (X-ray): Method 209 of MIL-STD-202.

(B) Thermal shock: Method 107 of MIL-STD-202, condition A, except step 3 shall be +125° +4, -0°C.

(B) Seal: Method 112 of MIL-STD-202, condition A (200-, 400-, and 600-volt units) or condition B (50-volt units); or (for liquid-filled units only) 1 hour at applicable high test temperature with no evidence of leakage.

Dielectric withstanding voltage (DWV): Method 301 of MIL-STD-202.
Terminal to terminal - 200 percent of rated voltage.
Terminals to case (when case is not a terminal) - 200 percent of rated voltage.
NOTE: For 100-percent inspection:
250 percent for not less than 5 seconds, or
200 percent for not less than 15 seconds.

Insulation resistance (IR): Method 302 of MIL-STD-202.
Terminal to terminal - See table III.
Terminals to case (when case is not a terminal) - 10,000 megohms, minimum.

Capacitance (Cap.): Method 305 of MIL-STD-202.

Dissipation factor (DF):
At 25°C - 1 percent.
At high test temperature - 2.0 percent.

Barometric pressure (reduced): Method 105 of MIL-STD-202, condition D (100,000 feet). 125 percent of rated voltage applied. See MIL-C-39022 for voltage limitations.

Vibration, high frequency: Method 204 of MIL-STD-202, condition D (20g peak). 50 percent of rated voltage applied.

Salt spray (corrosion): Method 101 of MIL-STD-202, condition B (48 hours).
Salt solution - 5 percent.

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TABLE I. Electrical characteristics, dimensions and dash numbers.

DC rated voltage	Capacitance value	Circuit diagram	Capacitance tolerance	Dimensions, nominal (inches) 1/						Dash number Failure rate level for (%/1,000 hr)				
				L .062	D .031	H .062	G .062	E .031	F .005	L(5.0)	M(1.0)	P(0.1)	R(0.01)	S(0.001)
Volts	µF		Percent (±)											
50	0.15	1	5	0.875	0.312	0.312	0.250	0.188	0.144	2001	2101	2201	2301	2401
50	0.15	1	10	0.875	0.312	0.312	0.250	0.188	0.144	2002	2102	2202	2302	2402
50	0.15	3	5	0.812	0.312	0.312	0.250	0.188	0.144	2003	2103	2203	2303	2403
50	0.15	3	10	0.812	0.312	0.312	0.250	0.188	0.144	2004	2104	2204	2304	2404
50	0.18	1	5	0.875	0.312	0.312	0.250	0.188	0.144	2005	2105	2205	2305	2405
50	0.18	1	10	0.875	0.312	0.312	0.250	0.188	0.144	2006	2106	2206	2306	2406
50	0.18	3	5	0.812	0.312	0.312	0.250	0.188	0.144	2007	2107	2207	2307	2407
50	0.18	3	10	0.812	0.312	0.312	0.250	0.188	0.144	2008	2108	2208	2308	2408
50	0.22	1	5	0.875	0.312	0.312	0.250	0.188	0.144	2009	2109	2209	2309	2409
50	0.22	1	10	0.875	0.312	0.312	0.250	0.188	0.144	2010	2110	2210	2310	2410
50	0.22	3	5	0.812	0.312	0.312	0.250	0.188	0.144	2011	2111	2211	2311	2411
50	0.22	3	10	0.812	0.312	0.312	0.250	0.188	0.144	2012	2112	2212	2312	2412
50	0.27	1	5	0.875	0.312	0.312	0.250	0.188	0.144	2013	2113	2213	2313	2413
50	0.27	1	10	0.875	0.312	0.312	0.250	0.188	0.144	2014	2114	2214	2314	2414
50	0.27	3	5	0.812	0.312	0.312	0.250	0.188	0.144	2015	2115	2215	2315	2415
50	0.27	3	10	0.812	0.312	0.312	0.250	0.188	0.144	2016	2116	2216	2316	2416
50	0.33	1	5	0.875	0.312	0.312	0.250	0.188	0.144	2017	2117	2217	2317	2417
50	0.33	1	10	0.875	0.312	0.312	0.250	0.188	0.144	2018	2118	2218	2318	2418
50	0.33	3	5	0.812	0.312	0.312	0.250	0.188	0.144	2019	2119	2219	2319	2419
50	0.33	3	10	0.812	0.312	0.312	0.250	0.188	0.144	2020	2120	2220	2320	2420
50	0.39	1	5	1.125	0.312	0.312	0.250	0.188	0.144	2021	2121	2221	2321	2421
50	0.39	1	10	1.125	0.312	0.312	0.250	0.188	0.144	2022	2122	2222	2322	2422
50	0.39	3	5	1.062	0.312	0.312	0.250	0.188	0.144	2023	2123	2223	2323	2423
50	0.39	3	10	1.062	0.312	0.312	0.250	0.188	0.144	2024	2124	2224	2324	2424
50	0.47	1	5	1.125	0.312	0.312	0.250	0.188	0.144	2025	2125	2225	2325	2425
50	0.47	1	10	1.125	0.312	0.312	0.250	0.188	0.144	2026	2126	2226	2326	2426
50	0.47	3	5	1.062	0.312	0.312	0.250	0.188	0.144	2027	2127	2227	2327	2427
50	0.47	3	10	1.062	0.312	0.312	0.250	0.188	0.144	2028	2128	2228	2328	2428
50	0.56	1	5	1.125	0.400	0.312	0.250	0.188	0.144	2029	2129	2229	2329	2429
50	0.56	1	10	1.125	0.400	0.312	0.250	0.188	0.144	2030	2130	2230	2330	2430
50	0.56	3	5	1.062	0.400	0.312	0.250	0.188	0.144	2031	2131	2231	2331	2431
50	0.56	3	10	1.062	0.400	0.312	0.250	0.188	0.144	2032	2132	2232	2332	2432
50	0.68	1	5	1.125	0.400	0.312	0.250	0.188	0.144	2033	2133	2233	2333	2433
50	0.68	1	10	1.125	0.400	0.312	0.250	0.188	0.144	2034	2134	2234	2334	2434
50	0.68	3	5	1.062	0.400	0.312	0.250	0.188	0.144	2035	2135	2235	2335	2435

See footnote at end of table.

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TABLE I. Electrical characteristics, dimensions and dash numbers - Continued.

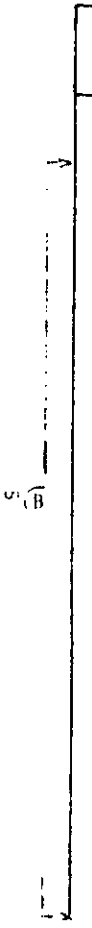
DC rated voltage	Capacitance value	Circuit diagram	Capacitance tolerance	Dimensions, nominal (inches) 1/						Dash number Failure rate level for (%/1,000 hr)				
				L .062	D .031	H .062	G .062	E .031	F .005	L(5.0)	M(1.0)	P(0.1)	R(0.01)	S(0.001)
Volts	µF		Percent (±)											
50	0.68	3	10	1.062	0.400	0.312	0.250	0.188	0.144	2036	2136	2236	2336	2436
50	0.82	1	5	1.125	0.400	0.312	0.250	0.188	0.144	2037	2137	2237	2337	2437
50	0.82	1	10	1.125	0.400	0.312	0.250	0.188	0.144	2038	2138	2238	2338	2438
50	0.82	3	5	1.062	0.400	0.312	0.250	0.188	0.144	2039	2139	2239	2339	2439
50	0.82	3	10	1.062	0.400	0.312	0.250	0.188	0.144	2040	2140	2240	2340	2440
50	1.0	1	5	1.125	0.400	0.312	0.250	0.188	0.144	2041	2141	2241	2341	2441
50	1.0	1	10	1.125	0.400	0.312	0.250	0.188	0.144	2042	2142	2242	2342	2442
50	1.0	3	5	1.062	0.400	0.312	0.250	0.188	0.144	2043	2143	2243	2343	2443
50	1.0	3	10	1.062	0.400	0.312	0.250	0.188	0.144	2044	2144	2244	2344	2444
50	1.2	1	5	1.375	0.400	0.312	0.250	0.188	0.144	2045	2145	2245	2345	2445
50	1.2	1	10	1.375	0.400	0.312	0.250	0.188	0.144	2046	2146	2246	2346	2446
50	1.2	3	5	1.312	0.400	0.312	0.250	0.188	0.144	2047	2147	2247	2347	2447
50	1.2	3	10	1.312	0.400	0.312	0.250	0.188	0.144	2048	2148	2248	2348	2448
50	1.5	1	5	1.125	0.500	0.312	0.250	0.188	0.144	2049	2149	2249	2349	2449
50	1.5	1	10	1.125	0.500	0.312	0.250	0.188	0.144	2050	2150	2250	2350	2450
50	1.5	3	5	1.062	0.500	0.312	0.250	0.188	0.144	2051	2151	2251	2351	2451
50	1.5	3	10	1.062	0.500	0.312	0.250	0.188	0.144	2052	2152	2252	2352	2452
50	1.8	1	5	1.375	0.500	0.312	0.250	0.188	0.144	2053	2153	2253	2353	2453
50	1.8	1	10	1.375	0.500	0.312	0.250	0.188	0.144	2054	2154	2254	2354	2454
50	1.8	3	5	1.312	0.500	0.312	0.250	0.188	0.144	2055	2155	2255	2355	2455
50	1.8	3	10	1.312	0.500	0.312	0.250	0.188	0.144	2056	2156	2256	2356	2456
50	2.2	1	5	1.125	0.562	0.438	0.500	0.250	0.156	2057	2157	2257	2357	2457
50	2.2	1	10	1.125	0.562	0.438	0.500	0.250	0.156	2058	2158	2258	2358	2458
50	2.2	3	5	1.062	0.562	0.438	0.500	0.250	0.156	2059	2159	2259	2359	2459
50	2.2	3	10	1.062	0.562	0.438	0.500	0.250	0.156	2060	2160	2260	2360	2460
50	2.7	1	5	1.375	0.562	0.438	0.500	0.250	0.156	2061	2161	2261	2361	2461
50	2.7	1	10	1.375	0.562	0.438	0.500	0.250	0.156	2062	2162	2262	2362	2462
50	2.7	3	5	1.312	0.562	0.438	0.500	0.250	0.156	2063	2163	2263	2363	2463
50	2.7	3	10	1.312	0.562	0.438	0.500	0.250	0.156	2064	2164	2264	2364	2464
50	3.3	1	5	1.375	0.562	0.438	0.500	0.250	0.156	2065	2165	2265	2365	2465
50	3.3	1	10	1.375	0.562	0.438	0.500	0.250	0.156	2066	2166	2266	2366	2466

See footnote at end of table.

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TABLE I. Electrical characteristics, dimensions and dash numbers - Continued.

DC rated voltage	Capacitance value	Circuit diagram	Capacitance tolerance	Dimensions, nominal (inches) 1/						Dash number Failure rate level for (%/1,000 hr)				
				L .062	D .031	H .062	G .062	E .031	F .005	L(5.0)	M(1.0)	P(0.1)	R(0.01)	S(0.001)
Volts	µF		Percent (±)											
50	3.3	3	5	1.312	0.562	0.438	0.500	0.250	0.156	2067	2167	2267	2367	2467
50	3.3	3	10	1.312	0.562	0.438	0.500	0.250	0.156	2068	2168	2268	2368	2468
50	3.9	1	5	1.625	0.562	0.438	0.500	0.250	0.156	2069	2169	2269	2369	2469
50	3.9	1	10	1.625	0.562	0.438	0.500	0.250	0.156	2070	2170	2270	2370	2470
50	3.9	3	5	1.562	0.562	0.438	0.500	0.250	0.156	2071	2171	2271	2371	2471
50	3.9	3	10	1.562	0.562	0.438	0.500	0.250	0.156	2072	2172	2272	2372	2472
50	4.7	1	5	1.750	0.562	0.438	0.500	0.250	0.156	2073	2173	2273	2373	2473
50	4.7	1	10	1.750	0.562	0.438	0.500	0.250	0.156	2074	2174	2274	2374	2474
50	4.7	3	5	1.688	0.562	0.438	0.500	0.250	0.156	2075	2175	2275	2375	2475
50	4.7	3	10	1.688	0.562	0.438	0.500	0.250	0.156	2076	2176	2276	2376	2476
50	5.6	1	5	1.875	0.562	0.438	0.500	0.250	0.156	2077	2177	2277	2377	2477
50	5.6	1	10	1.875	0.562	0.438	0.500	0.250	0.156	2078	2178	2278	2378	2478
50	5.6	3	5	1.812	0.562	0.438	0.500	0.250	0.156	2079	2179	2279	2379	2479
50	5.6	3	10	1.812	0.562	0.438	0.500	0.250	0.156	2080	2180	2280	2380	2480
50	6.8	1	5	1.625	0.670	0.438	0.500	0.250	0.156	2081	2181	2281	2381	2481
50	6.8	1	10	1.625	0.670	0.438	0.500	0.250	0.156	2082	2182	2282	2382	2482
50	6.8	3	5	1.562	0.670	0.438	0.500	0.250	0.156	2083	2183	2283	2383	2483
50	6.8	3	10	1.562	0.670	0.438	0.500	0.250	0.156	2084	2184	2284	2384	2484
50	8.2	1	5	1.875	0.670	0.438	0.500	0.250	0.156	2085	2185	2285	2385	2485
50	8.2	1	10	1.875	0.670	0.438	0.500	0.250	0.156	2086	2186	2286	2386	2486
50	8.2	3	5	1.812	0.670	0.438	0.500	0.250	0.156	2087	2187	2287	2387	2487
50	8.2	3	10	1.812	0.670	0.438	0.500	0.250	0.156	2088	2188	2288	2388	2488
50	10.0	1	5	1.875	0.750	0.438	0.500	0.250	0.156	2089	2189	2289	2389	2489
50	10.0	1	10	1.875	0.750	0.438	0.500	0.250	0.156	2090	2190	2290	2390	2490
50	10.0	3	5	1.812	0.750	0.438	0.500	0.250	0.156	2091	2191	2291	2391	2491
50	10.0	3	10	1.812	0.750	0.438	0.500	0.250	0.156	2092	2192	2292	2392	2492
50	12.0	1	5	2.375	0.750	0.438	0.500	0.250	0.156	2093	2193	2293	2393	2493
50	12.0	1	10	2.375	0.750	0.438	0.500	0.250	0.156	2094	2194	2294	2394	2494
50	12.0	3	5	2.312	0.750	0.438	0.500	0.250	0.156	2095	2195	2295	2395	2495
50	12.0	3	10	2.312	0.750	0.438	0.500	0.250	0.156	2096	2196	2296	2396	2496



See footnote at end of table.

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TABLE I. Electrical characteristics, dimensions and dash numbers - Continued.

DC rated voltage	Capacitance value	Circuit diagram	Capacitance tolerance	Dimensions, nominal (inches) 1/						Dash number Failure rate level for (%/1,000 hr)				
				L .062	D .031	H .062	G .062	E .031	F .005	L(5.0)	M(1.0)	P(0.1)	R(0.01)	S(0.001)
Volts	µF		Percent (±)											
200	0.10	1	10	0.844	0.312	0.312	0.250	0.188	0.144	1001	1121	1241	1361	1481
200	0.10	1	20	0.844	0.312	0.312	0.250	0.188	0.144	1002	1122	1242	1362	1482
200	0.10	3	10	0.781	0.312	0.312	0.250	0.188	0.144	1003	1123	1243	1363	1483
200	0.10	3	20	0.781	0.312	0.312	0.250	0.188	0.144	1004	1124	1244	1364	1484
200	0.22	1	10	1.125	0.312	0.312	0.250	0.188	0.144	1005	1125	1245	1365	1485
200	0.22	1	20	1.125	0.312	0.312	0.250	0.188	0.144	1006	1126	1246	1366	1486
200	0.22	3	10	1.062	0.312	0.312	0.250	0.188	0.144	1007	1127	1247	1367	1487
200	0.22	3	20	1.062	0.312	0.312	0.250	0.188	0.144	1008	1128	1248	1368	1488
200	0.47	1	10	1.125	0.400	0.312	0.250	0.188	0.144	1009	1129	1249	1369	1489
200	0.47	1	20	1.125	0.400	0.312	0.250	0.188	0.144	1010	1130	1250	1370	1490
200	0.47	3	10	1.062	0.400	0.312	0.250	0.188	0.144	1011	1131	1251	1371	1491
200	0.47	3	20	1.062	0.400	0.312	0.250	0.188	0.144	1012	1132	1252	1372	1492
200	1.0	1	10	1.125	0.562	0.438	0.500	0.250	0.156	1013	1133	1253	1373	1493
200	1.0	1	20	1.125	0.562	0.438	0.500	0.250	0.156	1014	1134	1254	1374	1494
200	1.0	3	10	1.062	0.562	0.438	0.500	0.250	0.156	1015	1135	1255	1375	1495
200	1.0	3	20	1.062	0.562	0.438	0.500	0.250	0.156	1016	1136	1256	1376	1496
200	1.5	1	10	1.844	0.562	0.438	0.500	0.250	0.156	1017	1137	1257	1377	1497
200	1.5	1	20	1.844	0.562	0.438	0.500	0.250	0.156	1018	1138	1258	1378	1498
200	1.5	3	10	1.781	0.562	0.438	0.500	0.250	0.156	1019	1139	1259	1379	1499
200	1.5	3	20	1.781	0.562	0.438	0.500	0.250	0.156	1020	1140	1260	1380	1500
200	2.2	1	10	1.844	0.562	0.438	0.500	0.250	0.156	1021	1141	1261	1381	1501
200	2.2	1	20	1.844	0.562	0.438	0.500	0.250	0.156	1022	1142	1262	1382	1502
200	2.2	3	10	1.781	0.562	0.438	0.500	0.250	0.156	1023	1143	1263	1383	1503
200	2.2	3	20	1.781	0.562	0.438	0.500	0.250	0.156	1024	1144	1264	1384	1504
200	3.3	1	10	1.875	0.670	0.438	0.500	0.250	0.156	1025	1145	1265	1385	1505
200	3.3	1	20	1.875	0.670	0.438	0.500	0.250	0.156	1026	1146	1266	1386	1506
200	3.3	3	10	1.812	0.670	0.438	0.500	0.250	0.156	1027	1147	1267	1387	1507
200	3.3	3	20	1.812	0.670	0.438	0.500	0.250	0.156	1028	1148	1268	1388	1508
200	4.7	1	10	1.875	1.000	0.438	0.500	0.250	0.156	1029	1149	1269	1389	1509
200	4.7	1	20	1.875	1.000	0.438	0.500	0.250	0.156	1030	1150	1270	1390	1510

See footnote at end of table.

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TABLE I. Electrical characteristics, dimensions and dash numbers - Continued.

DC rated voltage	Capacitance value	Circuit diagram	Capacitance tolerance	Dimensions, nominal (inches) 1/						Dash number Failure rate level for (%/1,000 hr)				
				L .062	D .031	H .062	G .062	E .031	F .005	L(5.0)	M(1.0)	P(0.1)	R(0.01)	S(0.001)
Volts	µF		Percent (±)											
200	4.7	3	10	1.812	1.000	0.438	0.500	0.250	0.156	1031	1151	1271	1391	1511
200	4.7	3	20	1.812	1.000	0.438	0.500	0.250	0.156	1032	1152	1272	1392	1512
200	6.8	1	10	1.875	1.000	0.438	0.500	0.250	0.156	1033	1153	1273	1393	1513
200	6.8	1	20	1.875	1.000	0.438	0.500	0.250	0.156	1034	1154	1274	1394	1514
200	6.8	3	10	1.812	1.000	0.438	0.500	0.250	0.156	1035	1155	1275	1395	1515
200	6.8	3	20	1.812	1.000	0.438	0.500	0.250	0.156	1036	1156	1276	1396	1516
200	10.0	1	10	2.375	1.000	0.438	0.500	0.250	0.156	1037	1157	1277	1397	1517
200	10.0	1	20	2.375	1.000	0.438	0.500	0.250	0.156	1038	1158	1278	1398	1518
200	10.0	3	10	2.312	1.000	0.438	0.500	0.250	0.156	1039	1159	1279	1399	1519
200	10.0	3	20	2.312	1.000	0.438	0.500	0.250	0.156	1040	1160	1280	1400	1520
200	12.0	1	10	2.625	1.000	0.438	0.500	0.250	0.156	1041	1161	1281	1401	1521
200	12.0	1	20	2.625	1.000	0.438	0.500	0.250	0.156	1042	1162	1282	1402	1522
200	12.0	3	10	2.562	1.000	0.438	0.500	0.250	0.156	1043	1163	1283	1403	1523
200	12.0	3	20	2.562	1.000	0.438	0.500	0.250	0.156	1044	1164	1284	1404	1524
400	0.047	1	10	1.125	0.312	0.312	0.250	0.188	0.144	1045	1165	1285	1405	1525
400	0.047	1	20	1.125	0.312	0.312	0.250	0.188	0.144	1046	1166	1286	1406	1526
400	0.047	3	10	1.062	0.312	0.312	0.250	0.188	0.144	1047	1167	1287	1407	1527
400	0.047	3	20	1.062	0.312	0.312	0.250	0.188	0.144	1048	1168	1288	1408	1528
400	0.10	1	10	1.125	0.400	0.312	0.250	0.188	0.144	1049	1169	1289	1409	1529
400	0.10	1	20	1.125	0.400	0.312	0.250	0.188	0.144	1050	1170	1290	1410	1530
400	0.10	3	10	1.062	0.400	0.312	0.250	0.188	0.144	1051	1171	1291	1411	1531
400	0.10	3	20	1.062	0.400	0.312	0.250	0.188	0.144	1052	1172	1292	1412	1532
400	0.22	1	10	1.125	0.562	0.438	0.500	0.250	0.156	1053	1173	1293	1413	1533
400	0.22	1	20	1.125	0.562	0.438	0.500	0.250	0.156	1054	1174	1294	1414	1534
400	0.22	3	10	1.062	0.562	0.438	0.500	0.250	0.156	1055	1175	1295	1415	1535
400	0.22	3	20	1.062	0.562	0.438	0.500	0.250	0.156	1056	1176	1296	1416	1536
400	0.47	1	10	1.625	0.562	0.438	0.500	0.250	0.156	1057	1177	1297	1417	1537
400	0.47	1	20	1.625	0.562	0.438	0.500	0.250	0.156	1058	1178	1298	1418	1538
400	0.47	3	10	1.562	0.562	0.438	0.500	0.250	0.156	1059	1179	1299	1419	1539
400	0.47	3	20	1.562	0.562	0.438	0.500	0.250	0.156	1060	1180	1300	1420	1540

See footnote at end of table.

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TABLE I. Electrical characteristics, dimensions and dash numbers - Continued.

DC rated voltage	Capacitance value	Circuit diagram	Capacitance tolerance	Dimensions, nominal (inches) 1/						Dash number Failure rate level for (%/1,000 hr)				
				L .062	D .031	H .062	G .062	E .031	F .005	L(5.0)	M(1.0)	P(0.1)	R(0.01)	S(0.001)
Volts	µF		Percent (±)											
400	1.0	1	10	1.875	0.750	0.438	0.500	0.250	0.156	1061	1181	1301	1421	1541
400	1.0	1	20	1.875	0.750	0.438	0.500	0.250	0.156	1062	1182	1302	1422	1542
400	1.0	3	10	1.812	0.750	0.438	0.500	0.250	0.156	1063	1183	1303	1423	1543
400	1.0	3	20	1.812	0.750	0.438	0.500	0.250	0.156	1064	1184	1304	1424	1544
400	2.2	1	10	1.875	1.000	0.438	0.500	0.250	0.156	1065	1185	1305	1425	1545
400	2.2	1	20	1.875	1.000	0.438	0.500	0.250	0.156	1066	1186	1306	1426	1546
400	2.2	3	10	1.812	1.000	0.438	0.500	0.250	0.156	1067	1187	1307	1427	1547
400	2.2	3	20	1.812	1.000	0.438	0.500	0.250	0.156	1068	1188	1308	1428	1548
400	3.3	1	10	2.625	1.000	0.438	0.500	0.250	0.156	1069	1189	1309	1429	1549
400	3.3	1	20	2.625	1.000	0.438	0.500	0.250	0.156	1070	1190	1310	1430	1550
400	3.3	3	10	2.562	1.000	0.438	0.500	0.250	0.156	1071	1191	1311	1431	1551
400	3.3	3	20	2.562	1.000	0.438	0.500	0.250	0.156	1072	1192	1312	1432	1552
600	0.01	1	10	0.812	0.312	0.312	0.250	0.188	0.144	1073	1193	1313	1433	1553
600	0.01	1	20	0.812	0.312	0.312	0.250	0.188	0.144	1074	1194	1314	1434	1554
600	0.01	3	10	0.750	0.312	0.312	0.250	0.188	0.144	1075	1195	1315	1435	1555
600	0.01	3	20	0.750	0.312	0.312	0.250	0.188	0.144	1076	1196	1316	1436	1556
600	0.022	1	10	0.812	0.312	0.312	0.250	0.188	0.144	1077	1197	1317	1437	1557
600	0.022	1	20	0.812	0.312	0.312	0.250	0.188	0.144	1078	1198	1318	1438	1558
600	0.022	3	10	0.750	0.312	0.312	0.250	0.188	0.144	1079	1199	1319	1439	1559
600	0.022	3	20	0.750	0.312	0.312	0.250	0.188	0.144	1080	1200	1320	1440	1560
600	0.047	1	10	1.125	0.400	0.312	0.250	0.188	0.144	1081	1201	1321	1441	1561
600	0.047	1	20	1.125	0.400	0.312	0.250	0.188	0.144	1082	1202	1322	1442	1562
600	0.047	3	10	1.062	0.400	0.312	0.250	0.188	0.144	1083	1203	1323	1443	1563
600	0.047	3	20	1.062	0.400	0.312	0.250	0.188	0.144	1084	1204	1324	1444	1564
600	0.10	1	10	1.125	0.500	0.312	0.250	0.188	0.144	1085	1205	1325	1445	1565
600	0.10	1	20	1.125	0.500	0.312	0.250	0.188	0.144	1086	1206	1326	1446	1566
600	0.10	3	10	1.062	0.500	0.312	0.250	0.188	0.144	1087	1207	1327	1447	1567
600	0.10	3	20	1.062	0.500	0.312	0.250	0.188	0.144	1088	1208	1328	1448	1568
600	0.15	1	10	1.125	0.562	0.438	0.500	0.250	0.156	1109	1229	1349	1469	1589
600	0.15	1	20	1.125	0.562	0.438	0.500	0.250	0.156	1110	1230	1350	1470	1590

See footnote at end of table.

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TABLE I. Electrical characteristics, dimensions and dash numbers - Continued.

DC rated voltage	Capacitance value	Circuit diagram	Capacitance tolerance	Dimensions, nominal (inches) 1/						Dash number Failure rate level for (%/1,000 hr)				
				L .062	D .031	H .062	G .062	E .031	F .005	L(5.0)	M(1.0)	P(0.1)	R(0.01)	S(0.001)
Volts	µF		Percent (±)											
600	0.15	3	10	1.062	0.562	0.438	0.500	0.250	0.156	1111	1231	1351	1471	1591
600	0.15	3	20	1.062	0.562	0.438	0.500	0.250	0.156	1112	1232	1352	1472	1592
600	0.22	1	10	1.375	0.562	0.438	0.500	0.250	0.156	1089	1209	1329	1449	1569
600	0.22	1	20	1.375	0.562	0.438	0.500	0.250	0.156	1090	1210	1330	1450	1570
600	0.22	3	10	1.312	0.562	0.438	0.500	0.250	0.156	1091	1211	1331	1451	1571
600	0.22	3	20	1.312	0.562	0.438	0.500	0.250	0.156	1092	1212	1332	1452	1572
600	0.47	1	10	1.625	0.670	0.438	0.500	0.250	0.156	1093	1213	1333	1453	1573
600	0.47	1	20	1.625	0.670	0.438	0.500	0.250	0.156	1094	1214	1334	1454	1574
600	0.47	3	10	1.562	0.670	0.438	0.500	0.250	0.156	1095	1215	1335	1455	1575
600	0.47	3	20	1.562	0.670	0.438	0.500	0.250	0.156	1096	1216	1336	1456	1576
600	1.0	1	10	1.844	1.000	0.438	0.500	0.250	0.156	1097	1217	1337	1457	1577
600	1.0	1	20	1.844	1.000	0.438	0.500	0.250	0.156	1098	1218	1338	1458	1578
600	1.0	3	10	1.781	1.000	0.438	0.500	0.250	0.156	1099	1219	1339	1459	1579
600	1.0	3	20	1.781	1.000	0.438	0.500	0.250	0.156	1100	1220	1340	1460	1580
600	1.5	1	10	1.875	1.000	0.438	0.500	0.250	0.156	1101	1221	1341	1461	1581
600	1.5	1	20	1.875	1.000	0.438	0.500	0.250	0.156	1102	1222	1342	1462	1582
600	1.5	3	10	1.812	1.000	0.438	0.500	0.250	0.156	1103	1223	1343	1463	1583
600	1.5	3	20	1.812	1.000	0.438	0.500	0.250	0.156	1104	1224	1344	1464	1584
600	2.2	1	10	2.625	1.000	0.438	0.500	0.250	0.156	1105	1225	1345	1465	1585
600	2.2	1	20	2.625	1.000	0.438	0.500	0.250	0.156	1106	1226	1346	1166	1586
600	2.2	3	10	2.562	1.000	0.438	0.500	0.250	0.156	1107	1227	1347	1467	1587
600	2.2	3	20	2.562	1.000	0.438	0.500	0.250	0.156	1108	1228	1348	1468	1588

1/ Metric equivalents are given in table II.

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TABLE II. Millimeter equivalents of decimal inches. 1/

Inches	MM	Inches	MM	Inches	MM	Inches	MM
.005	0.13	.400	10.16	0.875	22.23	1.750	44.45
.031	0.79	.438	11.13	1.000	25.40	1.781	45.24
.062	1.57	.500	12.70	1.062	26.97	1.812	46.02
.144	3.66	.562	14.27	1.125	28.58	1.844	46.84
.156	3.96	.670	17.02	1.312	33.32	1.875	47.63
.172	4.37	.750	19.05	1.375	34.93	2.312	58.72
.188	4.78	.781	19.84	1.562	39.67	2.375	60.33
.250	6.35	.812	20.62	1.625	41.28	2.562	65.07
.312	7.92	.844	21.44	1.688	42.88	2.625	66.68

1/ Metric equivalents (to the nearest .01 millimeter) are given for general information only, and are based upon 1 inch = 25.4 mm.

TABLE III. Insulation resistance.

	At dc voltage of			
	(B) 50V	200V	400V	(B) 600V
In megohms:				
At +25° ±3°C (need not exceed) - - - - -	30,000	12,000	12,000	12,000
At +125° +4, -0°C (need not exceed)- - -	150	150	600	600
In megohms x microfarads (minimum):				
At +25° ±3°C - - - - - - - - - - - - - -	20,000	2,000	2,000	2,000
At +125° +4, -0°C - - - - - - - - - - -	10	10	40	40

Immersion: Method 104 of MIL-STD-202, condition C.
DWV:
Terminal to terminal - 200 percent of rated voltage.
Terminals to case (when case is not a terminal) - 200 percent of rated voltage.
IR:
Terminal to terminal - Not less than 50 percent of value specified in table III.
Terminals to case (when case is not a terminal) - 5,000 megohms, minimum.
Cap. - Within ±10 percent of initial value.
DF - Not more than 1.1 percent.

Solderability: Method 208 of MIL-STD-202.

Shock (specified pulse): Method 213 of MIL-STD-202, condition I.
50 percent of rated voltage applied.

Moisture resistance: Method 106 of MIL-STD-202.
DWV, IR, Cap., and DF - Same as for immersion.

Terminal strength: Method 211 of MIL-STD-202, conditions A and D.
Condition A - Applied force 5 pounds.
Condition D - 3 rotations of 360 degrees.

Low temperature and cap. change with temperature:
Low temperature: -55°+0, -3°C for 48 ±4 hours with rated voltage applied.
Cap. change with temperature: At -55° +0, -3°C - -10 percent, maximum.
At +125° +4, -0°C - +20 percent, maximum.

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Fault count: In accordance with MIL-C-39022.
At 125° +4, -0°C monitored for momentary breakdowns and periods of low IR.
Qualification - 2,000 +72, -0 hours; quality conformance (group C)
240 +48, -0 hours.
Breakdowns - Shall not exceed 6C or 1, whichever is greater (where C is the nominal capacitance in µF of the sample). Total counts for any single capacitor in a 24-hour period shall not exceed 4C or 1, whichever is greater.
Period of low IR - Not to exceed 2 minutes.
IR:
Terminal to terminal - Not less than 33.3 percent of initial requirement.
Terminals to case (when case is not a terminal) - 5,000 megohms, minimum.
Cap. - Within ±10 percent of initial measured value.

Fungus: Method 508 of MIL-STD-810.

Resistance to soldering heat: Method 210 of MIL-STD-202, condition B (260 ±5°C for 10 ±1 seconds).
IR - See table III.
Cap. - Within ±5 percent of initial value.
DF - Not more than 1.0 percent.

Life: Method 108 of MIL-STD-202.
Qualification:
Accelerated conditions: 140 percent of rated voltage for 2,000 +72, -0 hours.
Rated conditions: 100 percent of rated voltage for 10,000 +96, -0 hours.
DF (at +125° +4, -0°C) between 24 and 48 hours of test - 2 percent, maximum.
During last 48 hours of test - 2.5 percent, maximum.
IR:
Insulating sleeves - 100 megohms, minimum.
Terminal to terminal - Not less than 33.3 percent of initial requirement.
Terminals to case (when case is not a terminal) - 5,000 megohms, minimum.
Cap. - Within ±10 percent of initial value.
DF (at +25° ±3°C) after life - 1.25 percent.

Quality conformance: 240 hours at accelerated conditions and meet the requirements for qualification test.

Extended life:
Accelerated conditions: 2,000 +72, -0 hours.
Rated conditions: 10,000 +96, -0 hours.

Marking: In accordance with MIL-C-39022.
Part number: M39022/08- (dash number from table I).

Custodians:
Army - EL
Navy - EC
Air Force - 11

Review activities:
Army - EL
Navy -
Air Force - 11, 17, 80
DSA - ES

User activities:
Army -
Navy - AS, MC, OS, SH, CG
Air Force - 19

Preparing activity:
Navy - EC

Agent:
DSA - ES

(Project 5910-1124)

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(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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